EXPERIMENT NO: 12

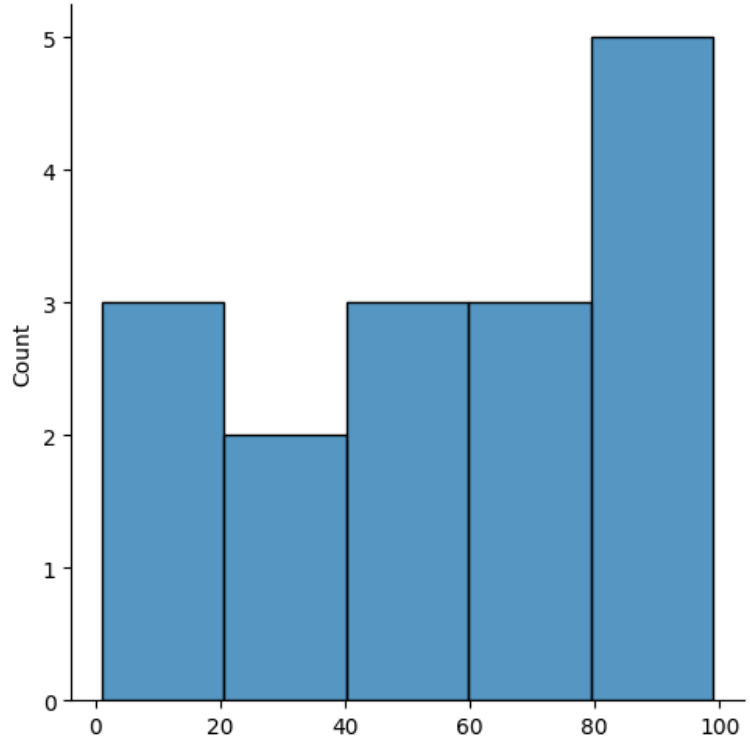
Experiment to detect outliers in a given dataset

Program:

```
import numpy as np
array = np.random.randint(1, 100, 16)
print(array)
print(array.mean())
print(np.percentile(array, 25))
print(np.percentile(array, 50))
print(np.percentile(array, 75))
print(np.percentile(array, 100))
def outDetection(array):
    sorted(array)
    Q1, Q3 = np.percentile(array, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
    ur = Q3 + (1.5 * IQR)
    return lr, ur
lr, ur = outDetection(array)
print(lr, ur)
import seaborn as sns
%matplotlib inline
sns.displot(array)
sns.distplot(array)
new_array = array[(array > lr) & (array < ur)]
print(new_array)
sns.displot(new_array)
lr1, ur1 = outDetection(new_array)
print(lr1, ur1)
final_array = new_array[(new_array > lr1) & (new_array < ur1)]
print(final_array)
sns.distplot(final_array)
```

Output:
[99 57 37 75 24 95 78 82 69 90 55 89 1 16 12 46]
57.8125
33.75
63.0
83.75
99.0
-41.25 158.75

<seaborn.axisgrid.FacetGrid at 0x7fae16eacad0>



/tmp/ipykernel_2265/1133588802.py:1: UserWarning:

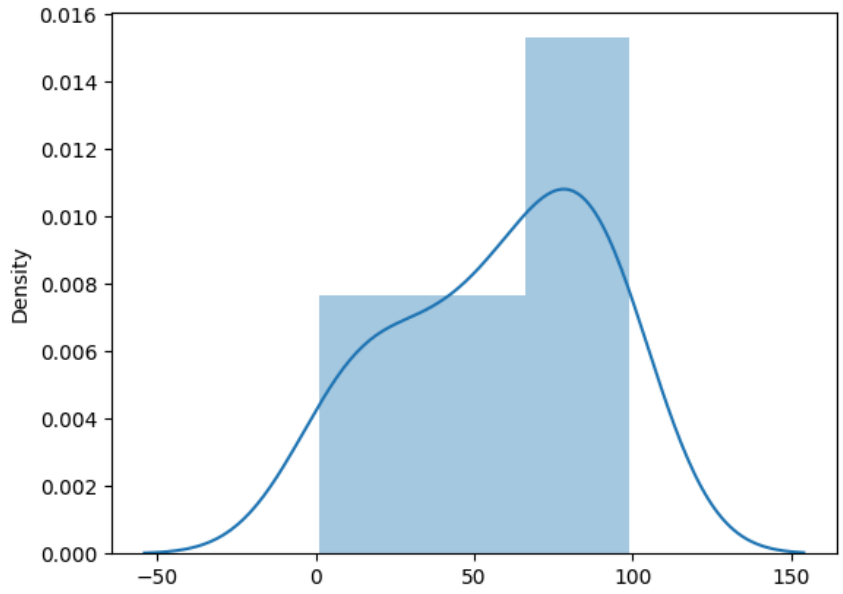
`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

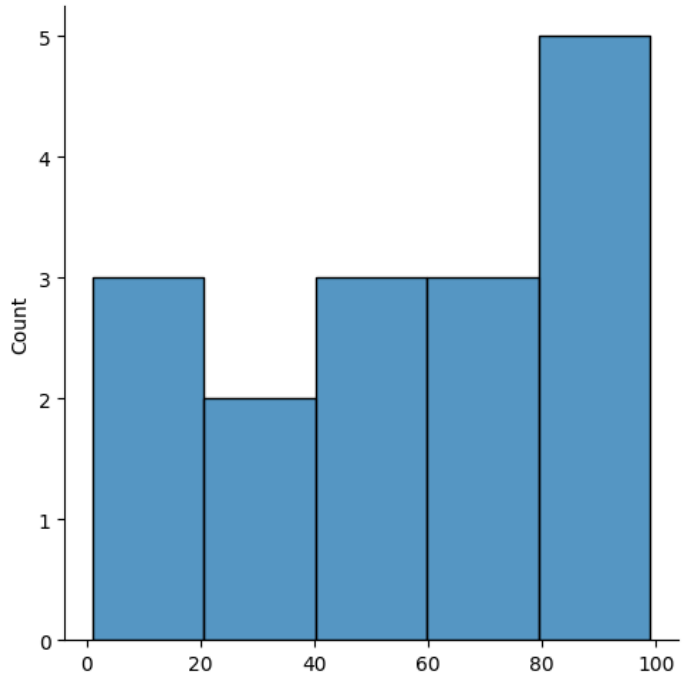
```
sns.distplot(array)
```

```
<Axes: ylabel='Density'>
```




```
[99 57 37 75 24 95 78 82 69 90 55 89  1 16 12 46]
```

```
<seaborn.axisgrid.FacetGrid at 0x7fae14ac3d90>
```



-41.25 158.75
[99 57 37 75 24 95 78 82 69 90 55 89 1 16 12 46]

/tmp/ipykernel_2265/2791332899.py:5: UserWarning:

`distplot` is a deprecated function and will be removed in seaborn v0.14.0.

Please adapt your code to use either `displot` (a figure-level function with similar flexibility) or `histplot` (an axes-level function for histograms).

For a guide to updating your code to use the new functions, please see
https://gist.github.com/mwaskom/de44147ed2974457ad6372750bbe5751

```
sns.distplot(final_array)
```

<Axes: ylabel='Density'>

